

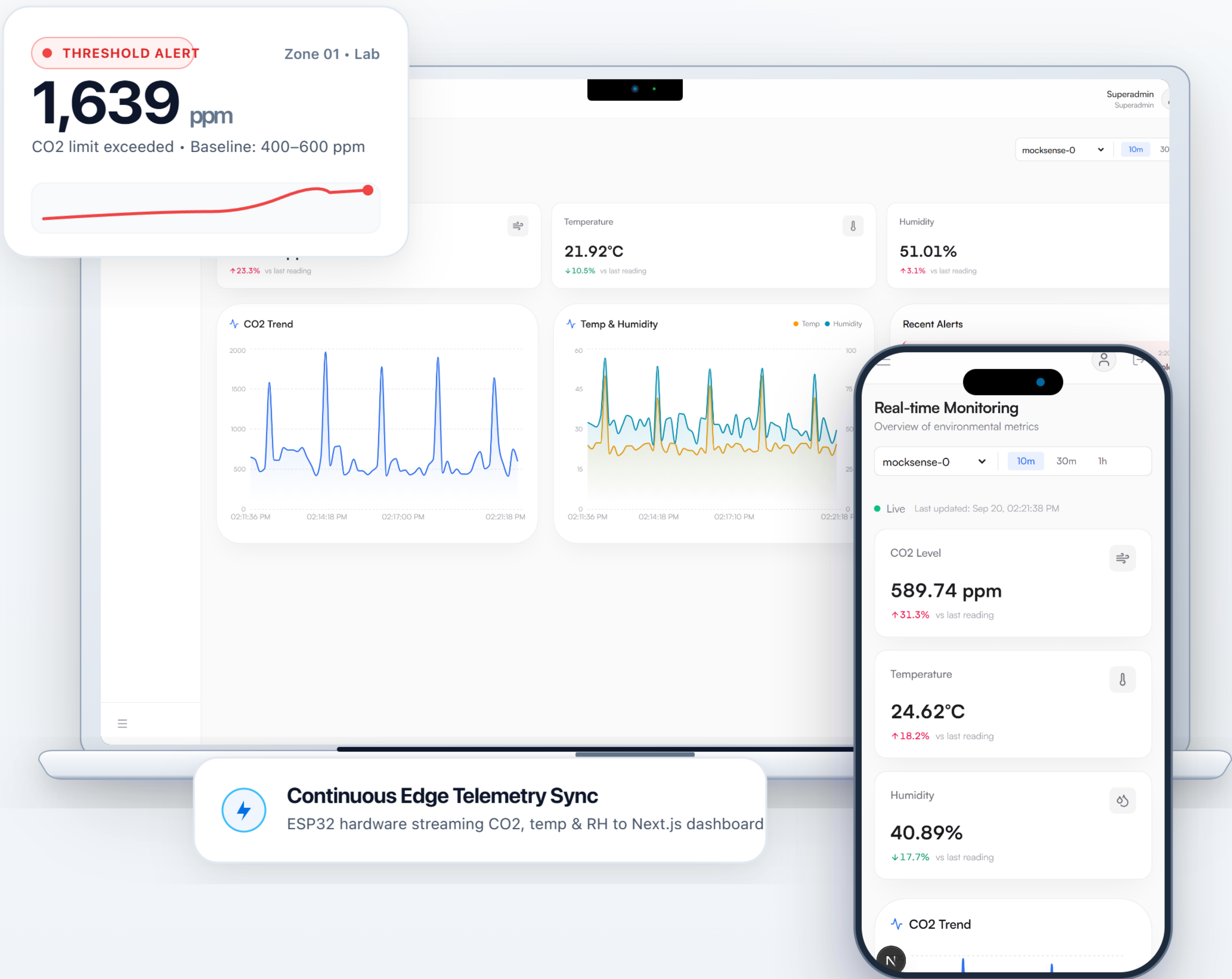


SENTINEL TELEMETRY

Environmental intelligence. Down to the molecule.

Continuous optical CO2, temperature, and humidity sensing.
Custom ESP32 edge telemetry streaming directly to your live console.

ESP32 HARDWARE NODES • NEXT.JS 16 APP ROUTER • POSTGRESQL ENGINE



SYSTEM PERFORMANCE SPECIFICATIONS

~10s

Hardware cadence.

Autonomous sensor nodes stream readings roughly every 10 seconds.

CO2 • H2

Multi-gas telemetry.

Optical NDIR CO2 sensor, MQ-2 combustible gas, ambient temp & humidity.

Instant

Threshold alarms.

Immediate visual alerts when hazard safety limits are breached.

ESP32 Firmware

Next.js 16 App Router

Neon Serverless Postgres

CSV Data Export



SENTINEL

Industrial Gas & Environmental Telemetry Platform

HARDWARE

ESP32 Sensor Nodes

- Optical NDIR CO2 Sensor
- MQ-2 Combustible Gas
- DHT22 Temp & Humidity
- Continuous WiFi Sync

WEB CONSOLE

Next.js 16 Dashboard

- Live Telemetry Waveforms
- Instant Threshold Alarms
- Multi-Zone Health Status
- Full-fidelity CSV Export

DATA ENGINE

Postgres & Drizzle

- Relational Sensor Records
- Type-Safe Database Schema
- Edge Ingestion REST API
- Time-Series History Log

ACTIVE DEPLOYMENTS

Zone 01: Lab & Testing

Zone 02: Production Floor

Zone 03: Storage & Air